Analysis Tutorial 12

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1 Introduction

In this tutorial, we will discuss metric spaces, differentiability of functions, and Riemann integrals.

2 Metric Spaces

Metric Spaces

A *metric space* is a pair (X, d) where X is a set and $d : X \times X \to [0, \infty)$ satisfies the following:

(1) $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$

(2) $d(x, y) = d(y, x)$

(3) $d(x, z) \leq d(x, y) + d(y, z)$

Open and Closed Sets

For $x \in X$ and $r > 0$, the *open ball* is

$$B(x, r) = \{y \in X : d(x, y) < r\}.$$

A set $U \subseteq X$ is open if for every $x \in U$ there exists $r > 0$ with $B(x, r) \subseteq U$. A set is closed if it contains all its limit points.

Convergence and Continuity

A sequence (x_n) converges to x if $d(x_n, x) \to 0$.

A function $f : (X, d_X) \to (Y, d_Y)$ is continuous at x if

$$\forall \varepsilon > 0 \; \exists \delta > 0 : d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Metric Transformation

Let (X, d) be a metric space and define $\qquad \delta(x, y) = \dfrac{d(x, y)}{1 + d(x, y)}$.

Proposition. δ is a metric on X.

3 Differentiability

1.1 Definition of the Derivative

Let $f : I \to \mathbb{R}$ be defined on an interval I and let $c \in I$. We say that f is *differentiable at* c if the limit

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. Equivalently,

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}.$$

1.2 Basic Properties

- If f is differentiable at c, then f is continuous at c.
- Linearity: $(\alpha f + \beta g)'(c) = \alpha f'(c) + \beta g'(c)$.
- Product Rule: $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.
- Quotient Rule:
$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$
- Chain Rule: $(g \circ f)'(c) = g'(f(c))f'(c)$.

1.3 Mean Value Theorem (MVT)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Exercises:

Problem 1. Let $f(x) = x^3$. Use the definition of the derivative to compute $f'(a)$.

Problem 2. Let f be differentiable on (a, b) with $f'(x) > 0$ for all x. Show that f is strictly increasing.

Problem 3. State the Mean Value Theorem and apply it to show:

$$|\sin x - \sin y| \leq |x - y|.$$

Problem 4. Suppose f is differentiable and $f'(x) = 0$ for all x. Show that f is constant.

Problem 5. Let f be differentiable on $(0, 1)$ and suppose $\lim_{x \to 0^+} f'(x)$ exists. Show that the limit

$$\lim_{x \to 0^+} \frac{f(x) - f(0)}{x}$$

exists.

4 Riemann Integrals

Riemann Integrability

A bounded function $f : [a, b] \to \mathbb{R}$ is *Riemann integrable* if

$$U(f, P) - L(f, P) \to 0 \quad \text{as} \quad \|P\| \to 0.$$

Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of the interval $[a, b]$, where

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The *norm* (or *mesh*) of the partition P is defined as

$$\|P\| = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Exercises:

Problem 1. Let

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Show that f is not Riemann integrable on any interval.

Problem 2. Prove that continuous functions are Riemann integrable.

Problem 3. Prove that if $f : [a, b] \to \mathbb{R}$ is bounded and continuous except at finitely many points, then f is Riemann integrable on $[a, b]$.